

A Novel Underwater Biomimetic Propulsion System Derived from Zebrafish Neurological Analysis

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Abstract—We propose a biomimetic propulsion system inspired by the spinal motor control mechanisms of zebrafish, allowing the submersible to adapt its propulsion strategy in real time based on real-time feedback from the environment. The design consists of two components: a neurological model and a topological model. The topological model is derived from Navier-Stokes equations to simulate fluid dynamics, coupled with topology optimization to optimize the shape and surface interactions of the submersible with water. To evaluate the effectiveness of the system, we tested the system under two control conditions: open-loop configurations and closed configurations. The experimental results demonstrated that the topology and topological models achieved significant improvements in energy efficiency and traversal time, compared to a control set-up. This design allows us to extend the operational range and duration of underwater vehicles, enabling more extensive and sustainable exploration of the ocean floor. This technology has far-reaching implications for ocean exploration, climate research, and underwater robotics. Moreover, the scalability of our design allows for potential applications in larger underwater vehicles as well as in multi-agent systems, such as swarms, where multiple underwater vehicles work together.

Index Terms—biomimetic submersible, undulatory swimming, computational fluid dynamics, neurological control model

I. INTRODUCTION

A. Background

Aquatic organisms have evolved over billions of years to optimize their locomotion for homeostasis, minimizing energy expenditure while maximizing propulsion. This is achieved through the continuous integration of sensory feedback and neuromuscular control, allowing them to fine-tune their movement in response to changes in environmental stimuli, such as water currents, pressure, and terrain. In particular, the biological propulsion systems present in aquatic organisms are remarkable due to the intricate signaling pathways that regulate their function. These pathways include spinal-based control for undulatory movements and peripheral motor circuits that govern tail fin oscillations and limb movements.

One such example is the zebrafish (*Danio rerio*), a small freshwater species native to Southeast Asia that has become a model organism in the study of neuromuscular control. Like many other aquatic organisms, the zebrafish maintains homeostasis by adjusting its movements in real time to conserve energy while maintaining functionality. This is due to its well-documented spinal cord circuitry and the relatively simple, yet highly effective, neural mechanisms it uses for locomotion. This ability to regulate movement through finely tuned neural circuits supports the use of biomimetic technology in the creation of energy-efficient propulsion systems. The Tuan Bui laboratory, for example, has developed computational models of spinal locomotor circuits in zebrafish, which replicate the developmental stages of their swimming behaviors, from early coiling to more complex behaviors such as burst swimming and beat-and-glide swimming [1].

In addition, zebrafish are capable of generating rhythmic swimming movements through central pattern generators (CPGs) in their spinal cord. CPGs are neural networks that can produce rhythmic outputs, controlling behaviors such as swimming without the need for sensory input. This ability to generate locomotion independently of external stimuli allows scientists to study intrinsic neural control of movement without interference from external sensory feedback, such as changes in water currents or terrain. The reliance of zebrafish on CPGs for swimming provides a unique opportunity to isolate and study the neural mechanisms that drive locomotion, offering valuable information for the development of a propulsion system inspired by their natural behavior.

B. Preliminary Investigation

Scientists and engineers have recognized the potential to design propulsion systems that mimic the neuromuscular pathways of aquatic organisms such as zebrafish, but a fully integrated propulsion system that combines fluid dynamics, neuromuscular control, and topological optimization has yet to be realized. Mitin et al. developed a robotic tuna-inspired



Fig. 1. Photo of a zebrafish by Massachusetts Institute of Technology Brain and Cognitive Sciences via <https://bcs.mit.edu/zebrafish>

fish, demonstrating how tail fin oscillations can improve speed and energy efficiency; however, this system lacked integration of sensory feedback [2]. Fluid dynamics advances, such as Evgrafov’s topology optimizations using Navier-Stokes equations, have enhanced flow characteristics, but did not address neuromuscular control [3]. Furthermore, Saputra et al. introduced a neurocognitive model for locomotion with sensory feedback, but their work focused on terrestrial environments [4].

Building on this previous research, we propose a propulsion system with two key components: a *neurological model* and a *topological model*. The neurological model draws inspiration from the synaptic activity of zebrafish, where peaks in electrocardiogram (ECG) data correlate with muscle contractions that generate propulsive forces (see Fig. 2). By analyzing and normalizing these data, we observed a relationship between electrical activity and motion efficiency. Translating this concept to robotics, we implemented a system that sends electrical “blips” to the engine at specific time steps, mimicking synaptic signals to achieve energy-efficient propulsion without compromising speed. This approach ensures that the submersible engine operates in a manner similar to the fine-tuned neuromuscular control of the zebrafish.

The topological model, on the other hand, determines the optimized hull shape of the submersible. Using Navier-Stokes equations to simulate fluid dynamics, coupled with topology optimization methods, we derived hull geometries that minimize hydrodynamic resistance. The integration of these two components allows the propulsion system to dynamically adapt to environmental factors, guided by real-time feedback through the Robot Operating System (ROS). Together, these components reduce energy consumption by aligning the propulsion mechanism with an optimized physical structure that is tailored to the fluid dynamics of its operating environment.

II. METHODOLOGY

A. Neurological Model

To create the neurological model, we used zebrafish ECG data collected by the Tuan Bui laboratory [1]. The raw ECG data were processed in MATLAB to identify the peaks

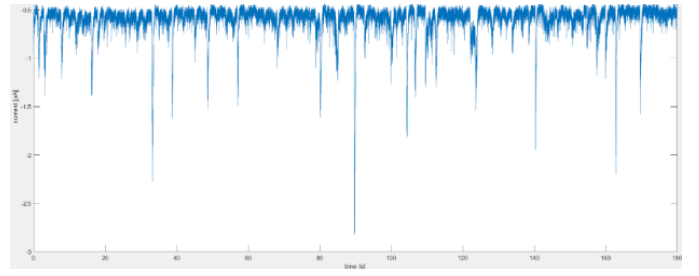


Fig. 2. Zebrafish ECG data shows peaks corresponding to spinal activity.

that represent synaptic activity. First, synaptic traces were normalized to ensure consistency throughout the data set. A structured array was reallocated to store validated synaptic events. For each trace, the signal was analyzed to detect peaks that exhibited the characteristic rapid rise and fall associated with synaptic activity. Peaks that met these criteria were recorded and further analyzed to quantify the frequency of synaptic events over time. This frequency, referred to as the synaptic value, represented the firing rate and served as the primary input for controlling the propulsion system.

B. Propulsion Framework With ROS

The synaptic value was transmitted to a motor control framework implemented using the Robot Operating System (ROS). This system dynamically adjusted the motor speed of the submersible according to modeled neural activity, analogous to how the CPGs of Zebrafish modulate muscle contractions. By integrating this feedback mechanism, the propulsion system achieved precise and efficient movement, reducing energy consumption without sacrificing performance.

The integration of a feedback system using ROS enables dynamic propulsion adjustments based on real-time environmental data. This system continuously monitors hydrodynamic parameters, such as velocity fields and pressure distributions, and adjusts the propulsion mechanism to maintain optimal efficiency. The zebrafish-inspired neural control model regulates these adjustments, drawing parallels with the CPG-driven swimming behavior of the organism.

C. Topological Model

To further optimize the propulsion system, we developed a topological model that minimized hydrodynamic resistance. This model incorporated principles from fluid dynamics, specifically the Navier-Stokes equations, to create vector fields that optimize the shape and surface interactions of the submersible with water. This process involved constructing a three-dimensional velocity field to simulate the effects of water pressure, viscosity, and flow patterns on the submersible’s performance.

D. Navier-Stokes Equations and Fluid Dynamics

The Navier-Stokes equations describe the motion of viscous fluid substances and form the basis of our CFD simulations.

The equations are expressed as:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (2)$$

where $\mathbf{u}$ represents the velocity field, ρ is the fluid density, p is the pressure, ν is the kinematic viscosity, and $\mathbf{f}$ denotes external forces. These equations are solved numerically using a computer-controlled model to simulate the flow around the submersible hull.

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1) *Component Form of the Navier-Stokes Equations:* The vector equation can be expressed in its component form as follows:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \quad (5)$$

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) \quad (6)$$

$$\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial z} + \nu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) \quad (7)$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad (8)$$

F. Topology Optimization for Hull Design

The hull shape is optimized using topology optimization, a mathematical approach to material distribution that minimizes a specified cost function subject to physical constraints. Here, the cost function represents the total hydrodynamic drag force, calculated as:

$$F_{\text{drag}} = \int_{\partial\Omega} \left(-p\mathbf{n} + \nu \frac{\partial \mathbf{u}}{\partial \mathbf{n}} \right) \cdot \mathbf{n} dA, \quad (9)$$

where $\partial\Omega$ is the surface of the hull, $\mathbf{n}$ is the outward normal vector, and A is the surface area. The optimization seeks to minimize F_{drag} .

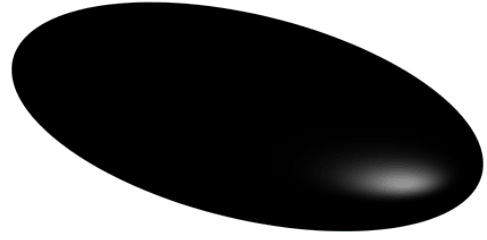


Fig. 3. The proposed submersible hull, constructed from vector field graphs based on Equations 3 and 4.

G. Functional Topological Model Definition

We define functional topological model “” as a three-dimensional surface structure designed to minimize the net force exerted by water on the object, quantified through power consumption and traversal time over a fixed distance (1 meter in our study). This force, $\mathbf{F}_{\text{net}}$, is computed as:

$$\mathbf{F}_{\text{net}} = \int_{\partial\Omega} \mathbf{T} \cdot \mathbf{n} dA, \quad (10)$$

where $\mathbf{T}$ is the stress tensor, including contributions from pressure and viscous forces:

$$\mathbf{T} = -p\mathbf{I} + \mu (\nabla \mathbf{u} + \nabla \mathbf{u}^T). \quad (11)$$

By iteratively refining the hull topology through CFD simulations, we identify a shape that achieves optimal hydrodynamic performance under the specified conditions (see Fig. 3).

III. EXPERIMENTAL DESIGN

A. Overview

The purpose of the experimental design was to evaluate the effectiveness of the topological model and biomimetic propulsion systems individually compared to a control setup. The experiments were carried out in a water tank with consistent conditions for all trials. Simulated waves were introduced into the tank to mimic the resistance and flow of water. Two key performance metrics were considered: traversal efficiency, measured by energy consumption in kilowatt hours, and hydrodynamic performance, assessed through traversal time across a fixed distance in seconds.

To isolate the impact of the neurological model, we employed two feedback systems: an open-loop and a closed-loop configuration. The open-loop system involved a fixed motor speed, independent of feedback or environmental conditions. In this configuration, the neurological model was not active, making the system dependent solely on reprogrammed propulsion patterns. In contrast, the closed-loop system incorporated feedback from the neurological model to dynamically adjust motor speed during trials. The neurological model interpreted synaptic data, sending electric blips to optimize propulsion in response to the simulated aquatic environment. This design mimicked the spinal motor control mechanisms observed in Zebrafish, allowing the submersible to adapt its propulsion strategy in real time. Similarly, we imposed a control on the topological model. The topological control is a cube as its

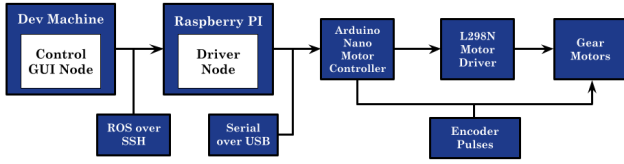


Fig. 4. Diagram of the circuit used for the experimental design.

regularity and symmetry make it easy to study the flow of fluid around it and how the system behaves when subjected to different conditions.

B. Experimental Controls

C. Procedure

The experimental procedure for this study evaluates both the propulsion system’s neurological model and the topological model’s design. The Propulsion Test focuses on comparing open-loop and closed-loop feedback systems, while the Model Design Test assesses the optimized hull shape against a control cube.

For the Propulsion Test, the first step is to 3D print the models and set up the necessary circuits, ensuring that the INA219 sensor, stepper motor, and feedback systems are properly configured. The experiment begins when both the control and experimental models are initiated simultaneously by uploading the ROS code. The systems are carefully monitored to avoid edge collisions and power consumption is recorded before and after each trial using the INA219 sensor. The experiment is repeated five times for both control and experimental configurations, and statistical analysis is performed to assess the significance of the results.

In the Model Design Test, the procedure is similar; however, the focus was on assessing the performance of the topological model. The models are again 3D printed and the circuits are configured according to figure 4. After ensuring equal mass between the control and experimental models, the ROS code is uploaded and both models are tested under identical conditions. The traversal time is recorded, and the experiment is repeated five times. As in the Propulsion Test, a Z test is used to determine statistical significance.

D. Setup

The submersibles were powered by LiPo batteries, with stepper motors controlled by ULN2003 drivers connected to an Arduino. The INA219 sensor was interfaced with a Raspberry Pi to monitor power consumption and motor commands were managed through ROS. For closed-loop control, feedback was continuously processed to adjust motor speed based on real-time synaptic data. Data logging was pre-installed in the system.

Before starting the trials, both the control and experimental models were calibrated to ensure equivalent starting conditions. Following the tests, power measurements were taken and the system was safely powered down to prepare for subsequent tests. To minimize external interference, the submersibles were

carefully aligned to avoid collisions with the edges of the tank during the trials. Environmental factors, such as water temperature and tank positioning, were monitored to maintain consistency in the trials.

The motor and system for both the control and the experimental design remained consistent throughout both experiments. Additionally, the mass of the wooden block was kept equivalent to that of the model in order to avoid confounding variables related to mass. For propulsion tests, the neurological model was always allowed to regulate motor speed based on synaptic values derived from zebrafish-inspired ECG data.

IV. RESULTS

In both open-loop and closed-loop tests, the biomimetic propulsion system demonstrated a statistically significant *improvement* in energy efficiency. Specifically, the z scores for the topological model and the propulsion system were -2.71 and -3.09, corresponding to p-values of 0.003 and 0.001, respectively. This confirmed that our biomimetic design was more energy-efficient than traditional propulsion systems. See Appendix A and Fig. 5. The closed-loop system, in particular, showed the greatest improvement in energy efficiency, as it was able to continuously adjust the propulsion force based on real-time feedback from the environment. This allowed the submersible to maintain a steady speed while minimizing energy expenditure in varying water conditions. The open-loop system, while still more efficient than traditional designs, was less adaptable to changing conditions and exhibited slightly higher energy consumption.

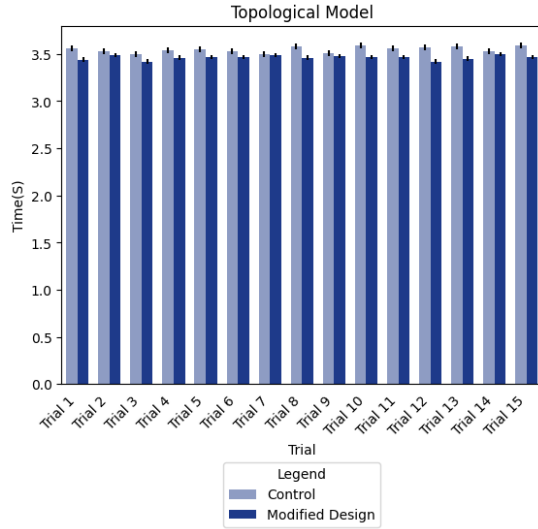
V. CONCLUSION

A. Implications

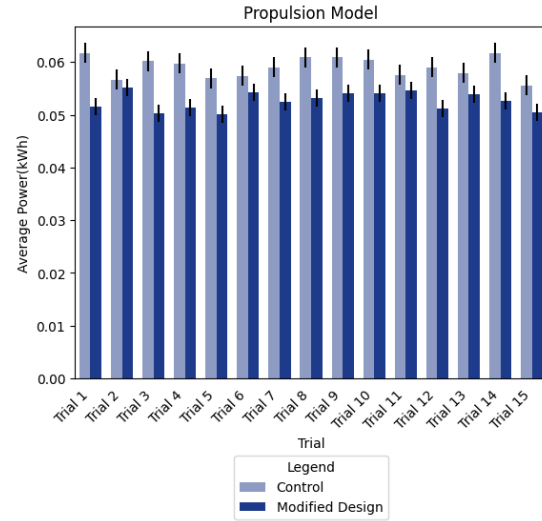
By reducing the energy requirements of submersibles, we can significantly extend their operational range and duration, allowing for more sustainable, long-term exploration of the ocean floor. This enhances our ability to monitor marine ecosystems and conduct extensive investigations of previously inaccessible regions. The ability to explore vast and remote areas of the ocean with minimal energy consumption will be particularly valuable for ocean topography, allowing more frequent and detailed mapping of the seafloor. This enhanced capability will provide critical data on underwater geological features, enabling a better understanding of ocean currents, tectonic activity, and habitat structures, all of which are vital for monitoring and mitigating the effects of climate change.

B. Future Work

Future work will focus on refining the submersible by pushing the boundaries of biohybridization and integrating additional sensors for advanced environmental monitoring. An area of particular interest is the integration of organoid research, where functional brain tissue is grown in controlled environments. The brains of fish, using neural cells, could be harnessed to control the operations of the submersible. This fusion of biological systems and robotics, driven by neural networks or organoid-derived pathways, could allow more



(a) Topological Model Results



(b) Neurological Model Results

Fig. 5. Results from both the topological and neurological models.

instinctual decision making, making autonomous underwater vehicles (AUVs) highly adaptive.

In addition to this bio-hybrid approach, the system will incorporate cutting-edge sensing technologies, including LiDAR, to facilitate high-precision real-time mapping and environmental monitoring. The ability of LiDAR to generate detailed topographical maps, in collaboration with global mapping initiatives, can significantly enhance submarine-like machine capabilities in seafloor exploration, marine ecosystem monitoring, and disaster response. Integrating these technologies with real-time environmental data processing will allow autonomous systems to dynamically adapt to complex underwater landscapes.

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APPENDIX A DATA TABLES

Test #2	Control	Modified Design
Trial 1	0.06167	0.0516
Trial 2	0.05667	0.05508
Trial 3	0.06015	0.05035
Trial 4	0.05969	0.05138
Trial 5	0.05688	0.05016
Trial 6	0.05733	0.05421
Trial 7	0.05904	0.05245
Trial 8	0.06092	0.05315
Trial 9	0.06092	0.05411
Trial 10	0.06044	0.05411
Trial 11	0.05752	0.05462
Trial 12	0.05902	0.05113
Trial 13	0.05789	0.05385
Trial 14	0.06173	0.05262
Trial 15	0.05554	0.05051
Average	0.0590273	0.05296
Standard	0.0019626	0.001620789
Z-score		-3.09151425
P-Value		<0.01

Fig. 6. Propulsion Test Results

Test #1	Control	Modified Design
Trial 1	3.56	3.44
Trial 2	3.53	3.49
Trial 3	3.5	3.42
Trial 4	3.54	3.46
Trial 5	3.55	3.47
Trial 6	3.53	3.47
Trial 7	3.5	3.49
Trial 8	3.58	3.46
Trial 9	3.51	3.48
Trial 10	3.59	3.47
Trial 11	3.56	3.47
Trial 12	3.57	3.42
Trial 13	3.58	3.45
Trial 14	3.53	3.5
Trial 15	3.59	3.47
Average	3.548	3.464
Standard Deviation	0.0309839	0.023543273
Z-score		-2.711088342
P-Value		0.034

Fig. 7. Model Test Results